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Midterm BGP Lab

CSCI 414 Platform-Based Development

Project Report

We created a compact network with four autonomous systems and two routes connecting their destination networks. We used Docker Compose to set up virtual links, FRRouting to exchange BGP routes, and Wireshark to analyze packets. We altered the preferred route, disabled its BGP partner, and examined whether the alternate route maintained connectivity.

GitHub repository: <https://github.com/benztranwot/BGP-Lab-CSCI-414>

I. Our topology

We linked left to top and bottom, and both routers to the right. The four connections use eBGP, with each router belonging to a separate AS. The figure displays the links and destination prefixes we advertised.

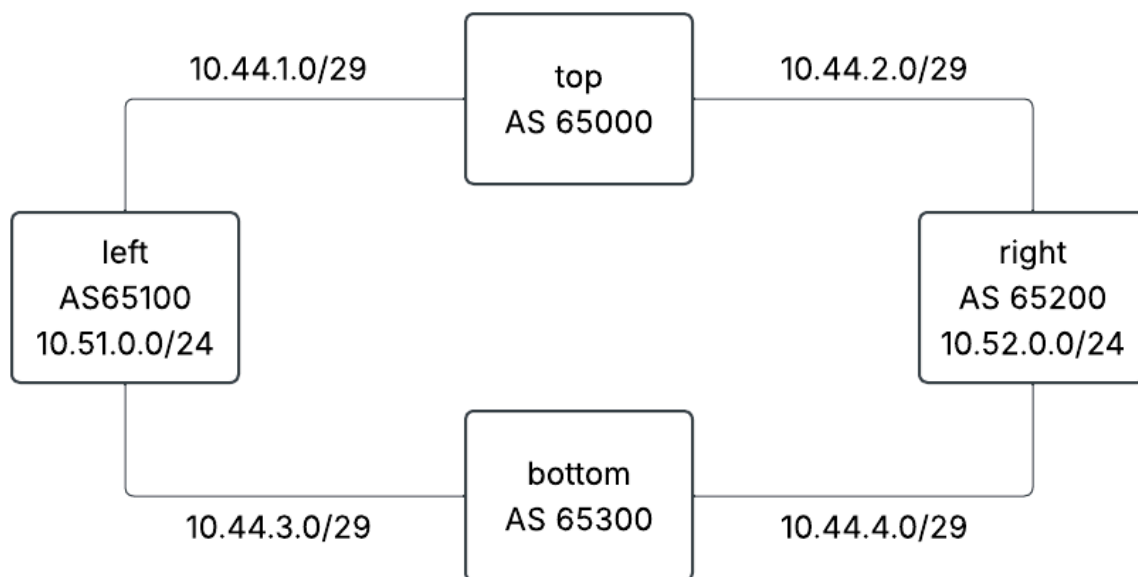


Diagram 1. Our four-AS topology. Each link uses a separate Docker bridge network.

Router	ASN	BGP router identifier	Advertised destination
left	65100	10.255.0.1	10.51.0.0/24
top	65000	10.255.0.2	None
bottom	65300	10.255.0.3	None
right	65200	10.255.0.4	10.52.0.0/24

We gave the left loopback interface 10.51.0.1/24 and the right one 10.52.0.1/24. The BGP network statements advertised the connected prefixes. We enabled Linux IP forwarding on all four containers. The table lists distinct BGP router identifiers for speakers and destination addresses.

Link	First endpoint	Second endpoint
left - top	left: 10.44.1.2	top: 10.44.1.3
top - right	top: 10.44.2.2	right: 10.44.2.3
left - bottom	left: 10.44.3.2	bottom: 10.44.3.3
bottom - right	bottom: 10.44.4.2	right: 10.44.4.3

II. Checking the baseline

We started by looking at the left BGP table and its two neighbors. Both routes to 10.52.0.0/24 were active. In this run, FRR chose the top and reported the best (Router ID). The top identifier, 10.255.0.2, is lower than the bottom's, 10.255.0.3. Both AS pathways include two AS numbers. We also confirmed that a ping sent from 10.51.0.1 could reach 10.52.0.1.

```
Windows PowerShell
containerd:
  Version: v2.3.5
  GitCommit: 1294c24a7da8e5a793ed378161673abe94118892
runc:
  Version: 1.5.1
  GitCommit: v1.5.1-0-g8f2685a4
docker-init:
  Version: 0.19.0
  GitCommit: de40ad0
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> docker compose up -d
[+] up 4/4
  ✓Container bgp-lab-right-1 Started 0.5s
  ✓Container bgp-lab-left-1 Started 0.8s
  ✓Container bgp-lab-top-1 Started 1.1s
  ✓Container bgp-lab-bottom-1 Started 1.3s
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> docker compose exec left vtysh -c "show bgp ipv4 unicast 10.52.0.0/24"
BGP routing table entry for 10.52.0.0/24, version 2
Paths: (2 available, best #1, table default)
  Advertised to peers:
    10.44.1.3 10.44.3.3
  65000 65200
    10.44.1.3 from 10.44.1.3 (10.255.0.2)
      Origin IGP, valid, external, best (Router ID)
      Last update: Thu Oct 1 22:19:04 2026
  65300 65200
    10.44.3.3 from 10.44.3.3 (10.255.0.3)
      Origin IGP, valid, external
      Last update: Thu Oct 1 22:19:04 2026
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab>
```

Figure 1. We had two candidate routes; the recorded baseline selected top

```
Windows PowerShell
65000 65200
  10.44.1.3 from 10.44.1.3 (10.255.0.2)
    Origin IGP, valid, external, best (Router ID)
    Last update: Thu Oct 1 22:19:04 2026
65300 65200
  10.44.3.3 from 10.44.3.3 (10.255.0.3)
    Origin IGP, valid, external
    Last update: Thu Oct 1 22:19:04 2026
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> docker compose exec left vtysh -c "show bgp ipv4 unicast summary"
BGP router identifier 10.255.0.1, local AS number 65100 VRF default vrf-id 0
BGP table version 2
RIB entries 2, using 320 bytes of memory
Peers 2, using 46 KiB of memory

Neighbor      V      AS  MsgRcvd  MsgSent  TblVer  InQ  OutQ  Up/Down  State/PfxRcd  PfxSnt Desc
10.44.1.3      4      65000      9        10        2    0    0 00:04:59      1          2 N/A
10.44.3.3      4      65300      9        10        2    0    0 00:04:59      1          2 N/A

Total number of neighbors 2
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> docker compose exec left ping -I 10.51.0.1 -c 3 10.52.0.1
PING 10.52.0.1 (10.52.0.1) from 10.51.0.1: 56 data bytes
64 bytes from 10.52.0.1: seq=0 ttl=63 time=0.159 ms
64 bytes from 10.52.0.1: seq=1 ttl=63 time=0.326 ms
64 bytes from 10.52.0.1: seq=2 ttl=63 time=0.336 ms

--- 10.52.0.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.159/0.273/0.336 ms
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab>
```

Figure 2. Both peers were established, and the baseline ping received 3/3 replies.

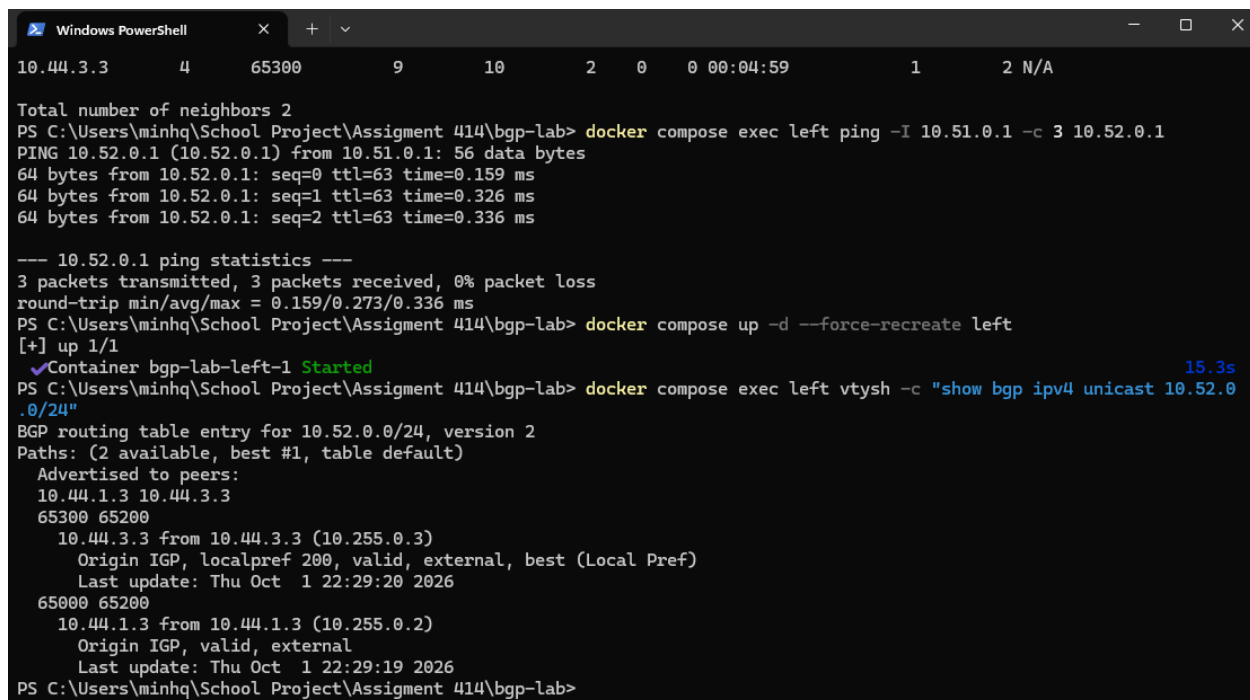
The value 1 under State/PfxRcd means each neighbor was established and supplied one prefix.

III. Making bottom the preferred route

We preferred left to bottom, despite the fact that both paths had the same AS-path length. We expected that giving the bottom's route a LOCAL_PREF of 200 would result in it outperforming the top's default value of 100. After adding an incoming route-map on the left, we saved the configuration and regenerated the container to load it.

```
route-map PREFER-BOTTOM permit 10
  set local-preference 200
!
router bgp 65100
  address-family ipv4 unicast
    neighbor 10.44.3.3 route-map PREFER-BOTTOM in
```

Route through	AS_PATH	LOCAL_PREF	After the change
top	65000 65200	100 (default)	Alternate
bottom	65300 65200	200	Selected



```
Windows PowerShell
10.44.3.3      4      65300      9      10      2      0      0 00:04:59      1      2 N/A

Total number of neighbors 2
PS C:\Users\minhq\School Project\Assigment 414\bgp-lab> docker compose exec left ping -I 10.51.0.1 -c 3 10.52.0.1
PING 10.52.0.1 (10.52.0.1) from 10.51.0.1: 56 data bytes
64 bytes from 10.52.0.1: seq=0 ttl=63 time=0.159 ms
64 bytes from 10.52.0.1: seq=1 ttl=63 time=0.326 ms
64 bytes from 10.52.0.1: seq=2 ttl=63 time=0.336 ms

--- 10.52.0.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.159/0.273/0.336 ms
PS C:\Users\minhq\School Project\Assigment 414\bgp-lab> docker compose up -d --force-recreate left
[+] up 1/1
✓Container bgp-lab-left-1 Started 15.3s
PS C:\Users\minhq\School Project\Assigment 414\bgp-lab> docker compose exec left vtysh -c "show bgp ipv4 unicast 10.52.0.0/24"
BGP routing table entry for 10.52.0.0/24, version 2
Paths: (2 available, best #1, table default)
  Advertised to peers:
    10.44.1.3 10.44.3.3
    65300 65200
    10.44.3.3 from 10.44.3.3 (10.255.0.3)
      Origin IGP, localpref 200, valid, external, best (Local Pref)
      Last update: Thu Oct 1 22:29:20 2026
    65000 65200
    10.44.1.3 from 10.44.1.3 (10.255.0.2)
      Origin IGP, valid, external
      Last update: Thu Oct 1 22:29:19 2026
PS C:\Users\minhq\School Project\Assigment 414\bgp-lab>
```

Figure 3. Bottom became best with localpref 200 and best (Local Pref).

The outcome matched our expectations. As LOCAL_PREF is tested before the router ID tie-breaker, this policy changed the left's option without altering either AS path. The value represents our routing choice within the AS, not bandwidth or physical distance.

IV. Disabling the preferred BGP peer

After running a continuous ping in a second terminal, we disconnected the left connection to the bottom and shut off neighbor 10.44.3.3. The administrative BGP peer termination did not affect the Docker link. Left removed the bottom candidate path and chose the top one that it already knew.

```

Last update: Thu Oct 1 22:29:19 2026
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> $failureTimer = [System.Diagnostics.Stopwatch]::StartNew()
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab>
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> docker compose exec -T left vtysh -c "configure terminal" -c "router bgp 65100" -c "neighbor 10.44.3.3 shutdown"
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab>
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> do {
>> $routes = docker compose exec -T left vtysh -c "show bgp ipv4 unicast"
>> $usingTop = ($routes -join "`n") -match '(?m)^s*\>s+10\.52\.0\.0/24\s+10\.44\.1\.3\b'
>> } while (-not $usingTop -and $failureTimer.Elapsed.TotalSeconds -lt 30)
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab>
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> $failureTimer.Stop()
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> $routes
BGP table version is 3, local router ID is 10.255.0.1, vrf id 0
Default local pref 100, local AS 65100
Status codes: s suppressed, d damped, h history, u unsorted, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*> 10.51.0.0/24      0.0.0.0              0         32768 i
*> 10.52.0.0/24      10.44.1.3              0         65000 65200 i

Displayed 2 routes and 2 total paths
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> "Top selected: $usingTop"
Top selected: True
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> "Alternate route observed after {0:N2} seconds" -f $failureTimer.Elapsed.TotalSeconds
Alternate route observed after 0.96 seconds
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> docker compose exec left vtysh -c "show bgp ipv4 unicast 10.52.0.0/24"
BGP routing table entry for 10.52.0.0/24, version 3
Paths: (1 available, best #1, table default)
  Advertised to peers:
    10.44.1.3
    65000 65200
    10.44.1.3 from 10.44.1.3 (10.255.0.2)
      Origin IGP, valid, external, best (First path received)
      Last update: Thu Oct 1 22:29:20 2026
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab>

```

Figure 4: Top became the selected route, first observed after 0.96 seconds.

What we checked	Observed result
Selected next hop	10.44.1.3 (top)
Selected AS_PATH	65000 65200
Candidate paths remaining	1
Time until our check observed top	0.96 seconds

Before sending the shutdown command, we began the stopwatch and examined the table until the top option was picked. The 0.96-second result includes Docker execution and checking overhead, making it an approximate upper bound for the observed switch. It does not measure the hold-timer delay caused by a silent link breakdown.

V. Watching traffic and restoring the peer

We received all 85 responses during the continuous ping test. There was no observed packet loss while we changed paths. Because the ping sends one packet each second, it may have missed a shorter interruption. The average round-trip time for this test was 0.310 ms.

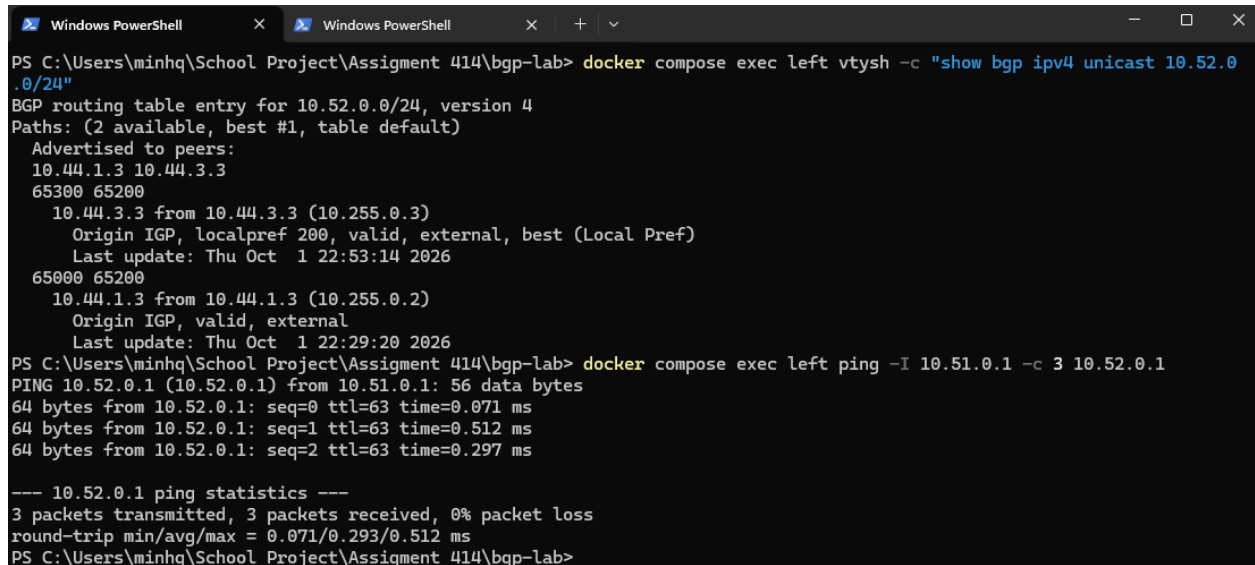
```

Windows PowerShell
64 bytes from 10.52.0.1: seq=67 ttl=63 time=0.075 ms
64 bytes from 10.52.0.1: seq=68 ttl=63 time=0.655 ms
64 bytes from 10.52.0.1: seq=69 ttl=63 time=0.268 ms
64 bytes from 10.52.0.1: seq=70 ttl=63 time=0.276 ms
64 bytes from 10.52.0.1: seq=71 ttl=63 time=0.298 ms
64 bytes from 10.52.0.1: seq=72 ttl=63 time=0.275 ms
64 bytes from 10.52.0.1: seq=73 ttl=63 time=0.330 ms
64 bytes from 10.52.0.1: seq=74 ttl=63 time=0.229 ms
64 bytes from 10.52.0.1: seq=75 ttl=63 time=0.279 ms
64 bytes from 10.52.0.1: seq=76 ttl=63 time=0.291 ms
64 bytes from 10.52.0.1: seq=77 ttl=63 time=0.268 ms
64 bytes from 10.52.0.1: seq=78 ttl=63 time=0.341 ms
64 bytes from 10.52.0.1: seq=79 ttl=63 time=0.285 ms
64 bytes from 10.52.0.1: seq=80 ttl=63 time=0.330 ms
64 bytes from 10.52.0.1: seq=81 ttl=63 time=0.336 ms
64 bytes from 10.52.0.1: seq=82 ttl=63 time=0.334 ms
64 bytes from 10.52.0.1: seq=83 ttl=63 time=0.332 ms
64 bytes from 10.52.0.1: seq=84 ttl=63 time=0.274 ms
^C
--- 10.52.0.1 ping statistics ---
85 packets transmitted, 85 packets received, 0% packet loss
round-trip min/avg/max = 0.066/0.310/1.107 ms
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab>

```

Figure 5. The monitoring ping received 85/85 replies with 0% observed loss.

We restored the peer with no neighbor 10.44.3.3 after it had been shut down. After the session resumed, the left had two options. Bottom won with LOCAL_PREF 200, and a final three-packet ping was successful.



```
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> docker compose exec left vtysh -c "show bgp ipv4 unicast 10.52.0.0/24"
BGP routing table entry for 10.52.0.0/24, version 4
Paths: (2 available, best #1, table default)
  Advertised to peers:
    10.44.1.3 10.44.3.3
    65300 65200
    10.44.3.3 from 10.44.3.3 (10.255.0.3)
      Origin IGP, localpref 200, valid, external, best (Local Pref)
      Last update: Thu Oct 1 22:53:14 2026
    65000 65200
    10.44.1.3 from 10.44.1.3 (10.255.0.2)
      Origin IGP, valid, external
      Last update: Thu Oct 1 22:29:20 2026
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab> docker compose exec left ping -I 10.51.0.1 -c 3 10.52.0.1
PING 10.52.0.1 (10.52.0.1) from 10.51.0.1: 56 data bytes
64 bytes from 10.52.0.1: seq=0 ttl=63 time=0.071 ms
64 bytes from 10.52.0.1: seq=1 ttl=63 time=0.512 ms
64 bytes from 10.52.0.1: seq=2 ttl=63 time=0.297 ms

--- 10.52.0.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.071/0.293/0.512 ms
PS C:\Users\minhq\School Project\Assignment 414\bgp-lab>
```

Figure 6. The preferred bottom path returned, and all three recovery pings succeeded.

VI. Recording the shutdown and reconnection

We used Netshoot to share the left network interfaces and tcpdump to record TCP port 179. To collect protocol messages, we repeated the shutdown and reconnection steps and opened bgp-evidence.pcap in Wireshark. The capture run was separate from the timed failure test.

```
Windows PowerShell x Windows PowerShell x + v
PS C:\Users\minhq\School Project\Assigment 414\bgp-lab> docker run --rm -it --name bgp-capture --network "container:$leftContainer" --cap-add NET_RAW --cap-add NET_ADMIN --mount "type=bind,source=$captureFolder,target=/captures" nicolaka/netshoot tcpdump -i any -nn -s 0 -U -w /captures/bgp-evidence.pcap "tcp port 179"
Unable to find image 'nicolaka/netshoot:latest' locally
latest: Pulling from nicolaka/netshoot
f4fcdd1672ed: Pull complete
72cec7a9d7fd: Pull complete
0f46943e91ef: Pull complete
dcf0aac248dc: Pull complete
fe519894e9d4: Pull complete
ff55a763fbcf: Pull complete
bea223344dda: Pull complete
ebb4e70d3b0a: Pull complete
55afa1ecc21d: Pull complete
0853a899d711: Pull complete
4f4fb700ef54: Pull complete
de71a8dba62c: Pull complete
6dfe05fc89ee: Pull complete
5a31b520f7cd: Pull complete
e831eb9a722d: Pull complete
dad86fcd0eb6: Download complete
Digest: sha256:b09d9b21381f47a79b3cbcb30da25266dc17186ea00ae65e99fdc51396f48e70
Status: Downloaded newer image for nicolaka/netshoot:latest
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
^C54 packets captured
54 packets received by filter
0 packets dropped by kernel
PS C:\Users\minhq\School Project\Assigment 414\bgp-lab>
```

Figure 7. We recorded 54 packets; tcpdump reported zero kernel capture drops.

During the outage, we noticed a NOTIFICATION running from left to right. It showed Cease (6), Hard Reset (9), and data 0602. According to RFC 8538, the original notification reason can be retained on a hard reset. Bytes 0602 identify Cease (6) and Administrative Shutdown (2).

The screenshot shows a Wireshark packet capture window titled 'bgp-evidence.pcap'. The packet list pane displays a series of BGP messages between 10.44.1.2 and 10.44.3.3. Frame 13 is selected, showing a BGP NOTIFICATION Message. The packet details pane for Frame 13 shows the BGP header and the notification message structure, including the Major error Code: Cease (6) and the Minor error Code (Cease): Hard Reset (9). The packet bytes pane shows the raw data of the packet, with the notification message starting at offset 0000.

No.	Time	Source	Destination	Protocol	Length	Info
5	6.867127	10.44.1.3	10.44.1.2	BGP	91	KEEPALIVE Message
7	59.989056	10.44.3.2	10.44.3.3	BGP	91	KEEPALIVE Message
8	59.996788	10.44.3.3	10.44.3.2	BGP	91	KEEPALIVE Message
10	66.852243	10.44.1.2	10.44.1.3	BGP	91	KEEPALIVE Message
11	66.855653	10.44.1.3	10.44.1.2	BGP	91	KEEPALIVE Message
13	93.136993	10.44.3.2	10.44.3.3	BGP	95	NOTIFICATION Message
17	93.199251	10.44.1.2	10.44.1.3	BGP	128	UPDATE Message
23	95.150745	10.44.3.3	10.44.3.2	BGP	192	OPEN Message
28	109.395968	10.44.3.2	10.44.3.3	BGP	192	OPEN Message
30	109.397131	10.44.3.3	10.44.3.2	BGP	192	OPEN Message
32	109.397221	10.44.3.3	10.44.3.2	BGP	91	KEEPALIVE Message
34	109.397239	10.44.3.2	10.44.3.3	BGP	91	KEEPALIVE Message
36	110.498166	10.44.3.2	10.44.3.3	BGP	206	UPDATE Message, UPDATE Message, UPDATE Message
38	110.548716	10.44.3.3	10.44.3.2	BGP	199	UPDATE Message, UPDATE Message, UPDATE Message
40	110.599141	10.44.3.2	10.44.3.3	BGP	128	UPDATE Message
42	110.599200	10.44.1.2	10.44.1.3	BGP	128	UPDATE Message
44	126.842462	10.44.1.2	10.44.1.3	BGP	91	KEEPALIVE Message
46	126.846052	10.44.1.3	10.44.1.2	BGP	91	KEEPALIVE Message
48	169.390724	10.44.3.2	10.44.3.3	BGP	91	KEEPALIVE Message
49	169.390870	10.44.3.3	10.44.3.2	BGP	91	KEEPALIVE Message
52	186.834634	10.44.1.2	10.44.1.3	BGP	91	KEEPALIVE Message

Frame 13: Packet, 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface 0
 Linux cooked capture v2
 Internet Protocol Version 4, Src: 10.44.3.2, Dst: 10.44.3.3
 Transmission Control Protocol, Src Port: 55212, Dst Port: 179, Seq: 39, Ack: 39, Len: 23
 Border Gateway Protocol - NOTIFICATION Message
 Marker: ffffffff
 Length: 23
 Type: NOTIFICATION Message (3)
 Major error Code: Cease (6)
 Minor error Code (Cease): Hard Reset (9)
 Data: 0602

0000 08 00 00 00 00 00 03 00 01 04 06 0e 04 a9 c4m...
 0010 65 8a 00 00 45 c0 00 4b 3a 94 40 00 01 06 23 fd e...E...K 1: @...#...
 0020 0a 2c 03 02 0a 2c 03 03 d7 ac 00 b3 d5 a8 6d 2f .,.....m/
 0030 8f ef 13 55 80 18 01 f5 1a 9a 00 00 01 01 08 0a ...U.....
 0040 6f f8 6e b3 8c 48 e6 a3 ff ff ff ff ff ff ff o-n-H.....
 0050 ff ff ff ff ff ff ff 00 17 03 06 09 06 02B.....

Major error Code (bgp.notify.major_error), 1 byte

Packets: 54 · Displayed: 25 (46.3%)

Profile: Default

Figure 8. Frame 13 records the Hard Reset and its administrative shutdown reason.

VII. Examining the BGP OPEN

We then evaluated an OPEN message sent by the bottom following reconnection. Bottom was detected as AS65300, with BGP identifier 10.255.0.3. The TCP source port was 179. The OPEN protocol includes a 180-second hold time for peers to negotiate session settings.

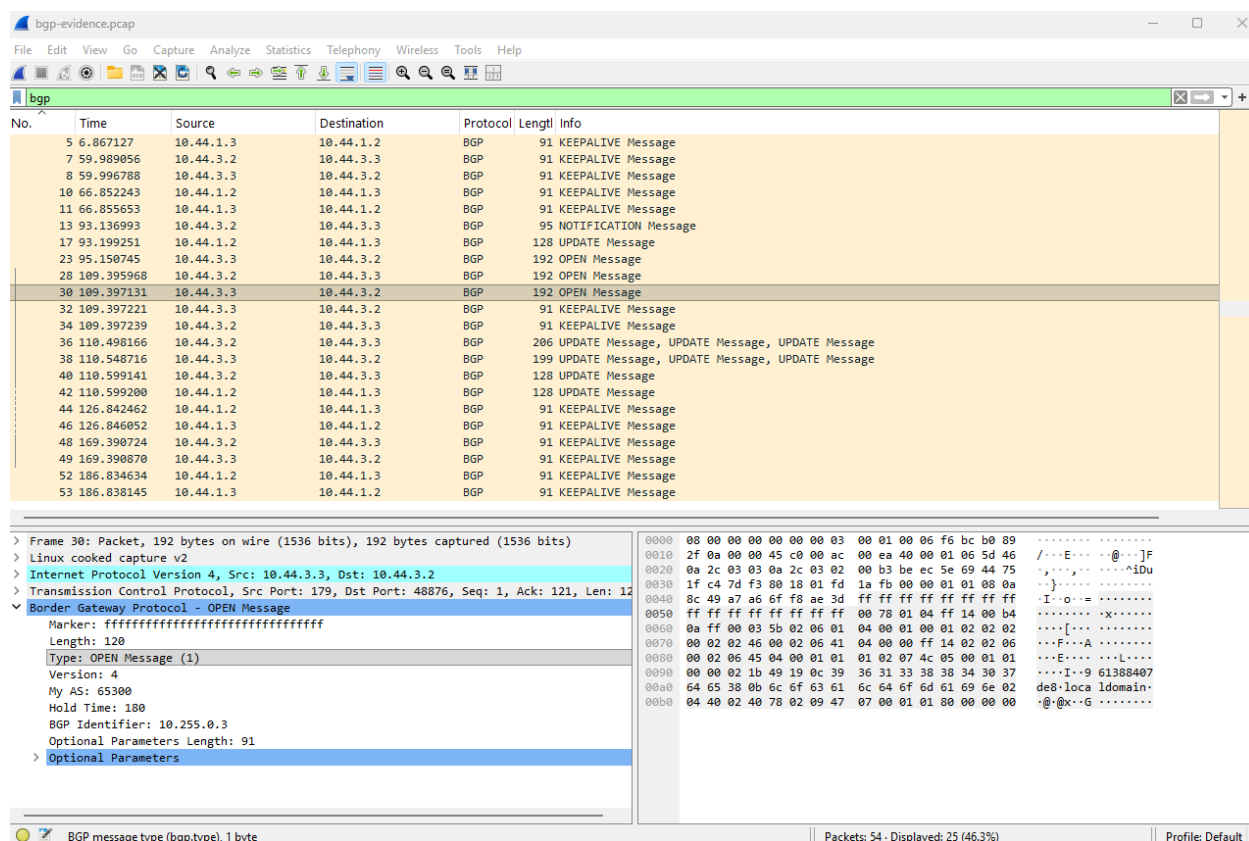


Figure 9. Frame 30 shows bottom's OPEN and the BGP TCP port.

Field	Recorded value
Source / destination	10.44.3.3 / 10.44.3.2
TCP source / destination port	179 / 48876
BGP type / version	OPEN (1) / version 4
My AS / BGP identifier	65300 / 10.255.0.3
Advertised hold time	180 seconds

We used this data to route the packet to the bottom router in our architecture. Seeing the OPEN after the shutdown confirmed the establishment of a fresh BGP session.

VIII. Following the route advertisement

We discovered an UPDATE from bottom to left that advertised the right's destination network. The AS_PATH was 65300:65200, and the NEXT_HOP was 10.44.3.3. These are the identical

path and next hop that we saw in the left routing table. We validated the destination prefix immediately in the saved grab, and the screenshot shows the NLRI tree collapsing.

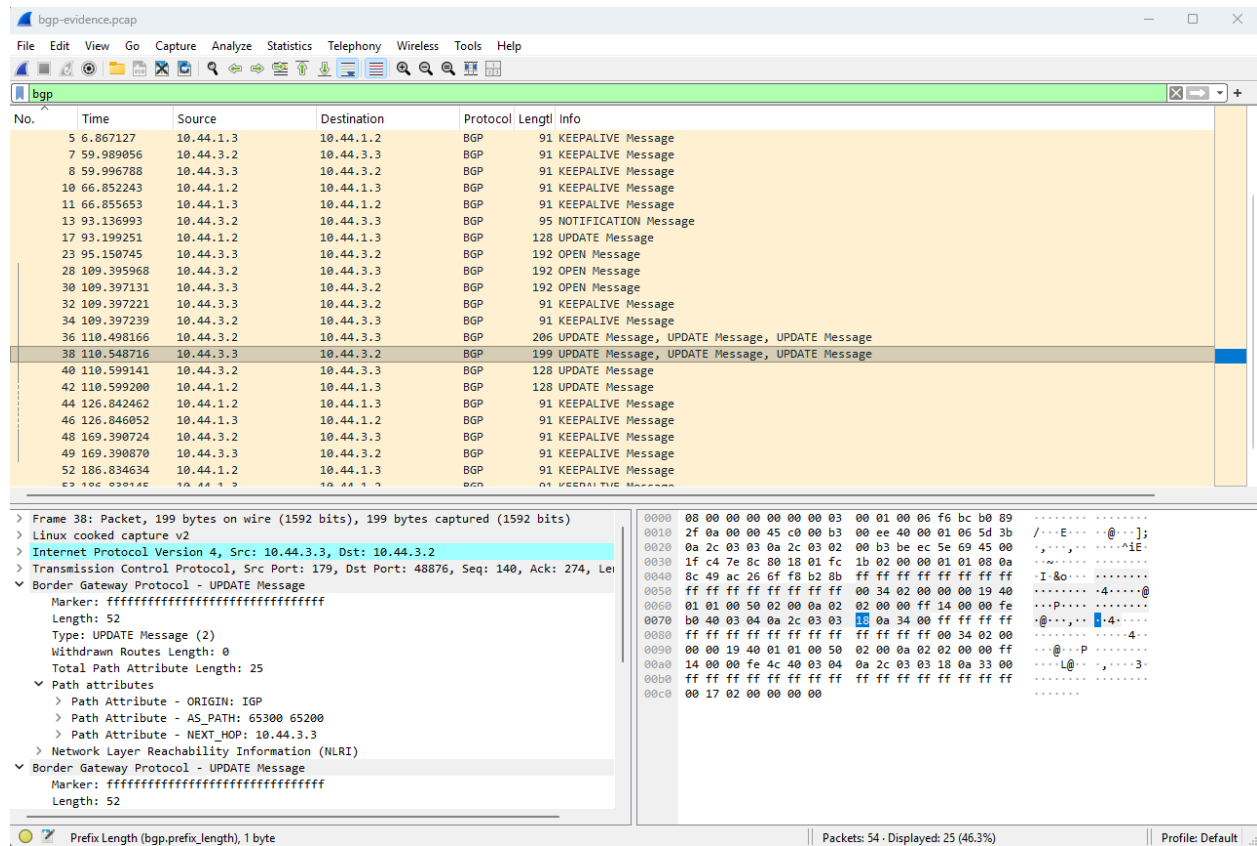


Figure 10. Frame 38 contains the route advertisement from bottom.

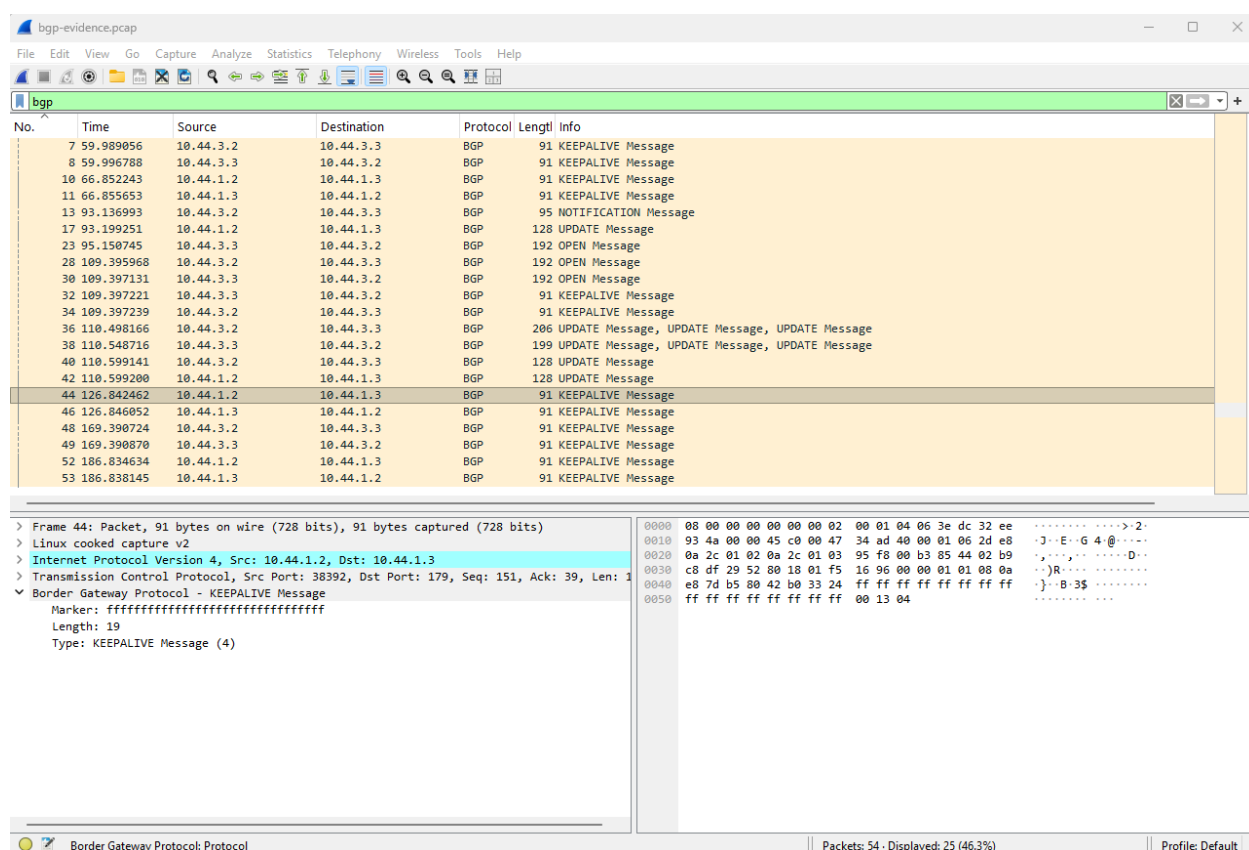
UPDATE field	Recorded value	What it tells us
NLRI	10.52.0.0/24	The destination network being advertised
AS_PATH	65300 65200	The AS sequence for the route
NEXT_HOP	10.44.3.3	Where left can send traffic next

We detected three BGP sections in this TCP frame because it had three different UPDATE messages. The initial IPv4 route exchange concluded with the first advertized right prefix and the final empty UPDATE. BGP operates on a TCP byte stream, allowing for several BGP messages in a single packet.

The inbound UPDATE did not include our LOCAL_PREF value of 200. We assigned the value locally on the left side of the incoming route-map after getting the bottom advertising. This separates the advertised path information from the policy used to select it.

IX. Checking KEEPALIVE and drawing conclusions

We also examined a KEEPALIVE that was transmitted from left to top. It was BGP type 4 with a 19-byte message length. A KEEPALIVE has simply the BGP header and keeps the session alive, but does not advertise any routes. The 19-byte value relates to the BGP message, not the whole collected frame.



No.	Time	Source	Destination	Protocol	Length	Info
7	59.989056	10.44.3.2	10.44.3.3	BGP	91	KEEPALIVE Message
8	59.996788	10.44.3.3	10.44.3.2	BGP	91	KEEPALIVE Message
10	66.852243	10.44.1.2	10.44.1.3	BGP	91	KEEPALIVE Message
11	66.855653	10.44.1.3	10.44.1.2	BGP	91	KEEPALIVE Message
13	93.136993	10.44.3.2	10.44.3.3	BGP	95	NOTIFICATION Message
17	93.199251	10.44.1.2	10.44.1.3	BGP	128	UPDATE Message
23	95.150745	10.44.3.3	10.44.3.2	BGP	192	OPEN Message
28	109.395968	10.44.3.2	10.44.3.3	BGP	192	OPEN Message
30	109.397131	10.44.3.3	10.44.3.2	BGP	192	OPEN Message
32	109.397221	10.44.3.3	10.44.3.2	BGP	91	KEEPALIVE Message
34	109.397239	10.44.3.2	10.44.3.3	BGP	91	KEEPALIVE Message
36	110.498166	10.44.3.2	10.44.3.3	BGP	206	UPDATE Message, UPDATE Message, UPDATE Message
38	110.548716	10.44.3.3	10.44.3.2	BGP	199	UPDATE Message, UPDATE Message, UPDATE Message
40	110.599141	10.44.3.2	10.44.3.3	BGP	128	UPDATE Message
42	110.599200	10.44.1.2	10.44.1.3	BGP	128	UPDATE Message
44	126.842462	10.44.1.2	10.44.1.3	BGP	91	KEEPALIVE Message
46	126.846852	10.44.1.3	10.44.1.2	BGP	91	KEEPALIVE Message
48	169.390724	10.44.3.2	10.44.3.3	BGP	91	KEEPALIVE Message
49	169.390870	10.44.3.3	10.44.3.2	BGP	91	KEEPALIVE Message
52	186.834634	10.44.1.2	10.44.1.3	BGP	91	KEEPALIVE Message
53	186.838145	10.44.1.3	10.44.1.2	BGP	91	KEEPALIVE Message

Frame 44: Packet, 91 bytes on wire (728 bits), 91 bytes captured (728 bits)	0000	08 00 00 00 00 00 02 00 01 04 06 3e dc 32 ee	> 2
> Linux cooked capture v2	0010	93 4a 00 00 45 c0 00 47 34 ad 40 00 01 06 2d e8	.J..E..G 4 @.....
> Internet Protocol Version 4, Src: 10.44.1.2, Dst: 10.44.1.3	0020	0a 2c 01 02 0a 2c 01 03 95 f8 00 b3 85 44 02 b9	.,.,.,.....D..
> Transmission Control Protocol, Src Port: 38392, Dst Port: 179, Seq: 151, Ack: 39, Len: 1	0030	c8 df 29 52 80 18 01 f5 16 96 00 00 01 01 08 0a	..)R.....
> Border Gateway Protocol - KEEPALIVE Message	0040	e8 7d b5 80 42 b0 33 24 ff ff ff ff ff ff ff	..}..B:3\$.....
Marker: ffffffff...	0050	ff ff ff ff ff ff ff 00 13 04	
Length: 19			
Type: KEEPALIVE Message (4)			

Figure 11. Frame 44 shows a 19-byte BGP KEEPALIVE.

X. Key Insights

We discovered that BGP route selection is dependent on both policy and the advertised AS path. Although both candidates had the same AS-path length, increasing the bottom's LOCAL_PREF resulted in a different winner. When we removed the favored peer and used the

existing alternate route, our monitoring pings continued to receive responses. Restoring the peer returned the policy-preferred route.

We related the routing tables to the messages that generated them: To start a session, use the OPEN command, UPDATE to update reachability and path properties, KEEPALIVE to keep it active, and NOTIFICATION to end it during a controlled shutdown. BGP's AS_PATH feature prevents loops by rejecting routes with the same AS as the router. Our findings show policy control and recovery in this four-AS lab.